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Region-resolved closure tests of the Planck PR4 polarization-angle calibration network and the degeneracy of birefringence with dust misalignment

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ABSTRACT

Cosmic birefringence searches are limited by the absolute calibration of detector polarization angles. Lonappan et al. (2026) introduced a birefringence-blind differential test on the eight Planck PR4 (NPIPE) detector-split maps: pairwise angle differences form a network whose consistency with the Minami–Komatsu (MK) calibration pattern can be checked without assuming foreground EB. We present an independent implementation on the public maps. The 28 pair angles close ($\chi^2 = 28.8$ for 21 degrees of freedom) and agree with an MK fit of the same maps ($\chi^2_{\rm diff} = 3.32$ for 7 dof, PTE 0.85; Pearson $r = 0.984$), with the largest pull at the same map the original analysis identified. We formulate the network as a cochain complex on the filled complete graph K_8: the common rotation is the gauge kernel of the coboundary, $H^1 = 0$, and the residual of the covariance-weighted fit is a weighted coexact component whose quadratic form equals the triangle-closure statistic. We then ask a question the global analysis cannot: is the calibration pattern the same across the sky? Splitting the misfit exactly into within-region closure and a spatial-stationarity term, with blinded, pre-registered thresholds, we find no evidence for spatial variation over twelve equal-area regions ($\chi^2 = 70.9$ for 77 dof, PTE 0.67), a test that detects a 0.1° instrument-like drift 90% of the time. A latitude-band partition shows a hint of variation on the dust-immune same-frequency edges that does not survive correction for the number of statistics reported. Finally, a coverage-validated neural posterior estimator over a family of dust-EB models shows that, once the dust misalignment ψ is free, β is degenerate with it (exactly so for a scale-independent misalignment): these channels measure $\beta - \psi_0 = -0.09^{+0.29}_{-0.28}$ deg, and β itself follows only from an assumed prior on ψ_0 ($\beta = -0.07^{+0.55}_{-0.52}$ deg for $\psi_0 \sim \mathcal{N}(0, 0.5^\circ)$).

Keywords: cosmic background radiation — polarization — methods: data analysis

1. INTRODUCTION

A parity-violating rotation of the CMB polarization plane by an angle β would convert E-mode into B-mode power and produce a nonzero EB correlation. Recent analyses of Planck, WMAP and ACT data report $\beta \approx 0.2^\circ$–0.4° (Minami & Komatsu 2020; Diego-Palazuelos et al. 2022; Eskilt & Komatsu 2022; ACT Collaboration 2025), but β is exactly degenerate with a common rotation of all detector angles. The MK method (Minami et al. 2019; Minami & Komatsu 2020) breaks the degeneracy by using Galactic foreground emission, which is rotated by the instrument but not by β, at the cost of assumptions about the intrinsic EB of dust (Clark et al. 2021).

Lonappan et al. (2026) proposed a complementary consistency test. For two maps i, j rotated by α_i, α_j, the antisymmetric and symmetric cross-spectra satisfy

$$C_\ell^{B_iE_j} - C_\ell^{E_iB_j} = \tan\big[2(\alpha_i - \alpha_j)\big]\,\big(C_\ell^{E_iE_j} + C_\ell^{B_iB_j}\big) \quad (1)$$

for any sky EE, BB and intrinsic EB, so the common rotation cancels. The 28 pair angles of the eight NPIPE detector-split maps reconstruct the relative calibration pattern, which can be compared with the MK pattern on the seven-dimensional differential subspace.

This paper (i) independently implements the test on the public maps and quantifies agreement, (ii) states the cochain structure of the network precisely, (iii) introduces a region-resolved extension that tests whether the calibration pattern is spatially stationary, with a dust-immune instrument control, (iv) quantifies with simulation-based inference what the same channels can say about β once the dust misalignment is free, and (v) reports a topological diagnostic of the temperature map against noise-realistic and phase-randomized nulls.

2. DATA AND SPECTRA

We use the eight PR4 NPIPE detector-split maps at $N_{\rm side} = 2048$ (100, 143, 217 and 353 GHz, splits A and

B; each file header records its `npipe6v20A/B` source) (Planck Collaboration 2020). The masks of Lonappan et al. (2026) are not public; we rebuild them from Planck Legacy Archive products: the union of the polarized point-source masks at 100–353 GHz times a CO(2–1) cut chosen to reach $f_{\rm sky} = 0.92$, and the same times the PLA Galactic mask whose combination lands nearest $f_{\rm sky} = 0.62$ (0.60). Spin-2 harmonic transforms use `ducc0` (Reinecke 2023); all 16×16 field cross-spectra are binned in 72 bins of width 20 over $51 \le \ell < 1491$. We divide pseudo-C_ℓ by $\langle w^2 \rangle$ rather than deconvolving mode coupling; pair angles are ratios and are insensitive to this, while for the MK theory term we apply a mask transfer function measured from ΛCDM simulations. Beam windows are the NPIPE QuickPol products.

3. THE NETWORK AS A COCHAIN COMPLEX

Let maps be the vertices of K_8, pairs its 28 edges and fill all 56 triangles. With $(d_0 x)_{ij} = x_i - x_j$ and $(d_1 w)_{ijk} = w_{ij} + w_{jk} - w_{ik}$, the pair-angle model is $\Delta\alpha = d_0\alpha + n$, so the incidence matrix of Lonappan et al. (2026) is the coboundary d_0, and $\ker d_0 = \mathrm{span}\{\mathbf{1}\}$ is the common rotation (the gauge freedom they note). Since $\mathrm{rank}\, d_0 = 7$ and $\mathrm{rank}\, d_1 = 21$, the first cohomology vanishes, $H^1 = 0$: no harmonic component exists, and anything outside $\mathrm{im}\, d_0$ is a failure of triangle closure.

The generalized least-squares fit with edge covariance C is orthogonal in the C^{-1} metric. Its residual r obeys $d_0^\top C^{-1} r = 0$, so $C^{-1} r \in \mathrm{im}\, d_1^\top$: r is a *covariance-weighted* coexact component, and need not lie in $\mathrm{im}\, d_1^\top$ itself unless C preserves that subspace. The metric-free closures $d_1 \Delta\alpha$ carry the same information: their quadratic form $c^\top (d_1 C d_1^\top)^+ c$ equals the GLS residual χ^2 with 21 degrees of freedom. This graph "curl" is a calibration diagnostic, unrelated to CMB B-modes. All numerical kernels are verified in a network-less sandbox with hash-chained receipts (Appendix A).

4. REPRODUCTION

Pair angles come from a reweighted least-squares fit of Eq. 1 across bins with Gaussian bandpower covariance propagated to the 28×28 pair covariance; their calibration was validated on injected synthetic skies (pull standard deviation 0.90). The gauge-fixed network fit gives $\chi^2 = 28.8$ for 21 dof (PTE 0.12) at $f_{\rm sky} = 0.92$ and 30.7 (PTE 0.08) at $f_{\rm sky} = 0.60$. A numpy port of the MK likelihood of Lonappan et al. (2026), with four dust-EB amplitudes, yields the per-map pattern of Fig. 1. Keeping the network–MK cross-covariance from 2,000 bootstrap draws of the bandpowers, $\chi^2_{\rm diff} = 3.32$ for 7 dof (PTE 0.85), compared with 8.18 (PTE 0.32) in Lonappan et al. (2026); the largest pull is $143B$ ($+0.70\sigma$), the map they also single out. Dust C_ℓ^{EE}/C_ℓ^{BB} at $\ell < 300$ is 1.41, 1.40 and 1.40 at 143, 217 and 353 GHz.

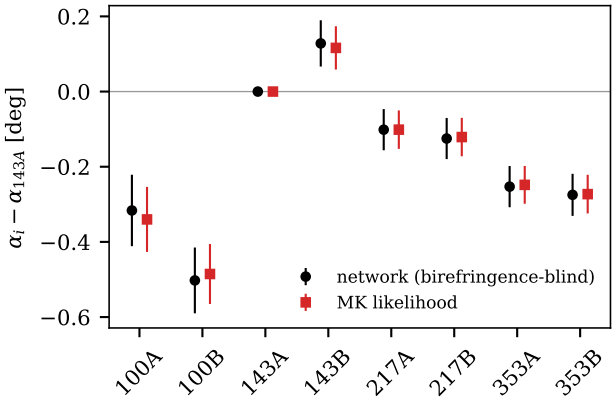


Figure 1. Per-map calibration pattern (gauge $\alpha_{143A} = 0$) from the birefringence-blind network and from the MK likelihood on the same maps.

The MK birefringence angle is $\beta = 0.25^\circ \pm 0.23^\circ$ (a profile likelihood including $\ln\det C$ gives 0.25 ± 0.23), consistent with $0.37^\circ \pm 0.11^\circ$ (Lonappan et al. 2026) but with about twice the uncertainty. It depends on the dust-EB model: -0.08 ± 0.14 with no dust EB and 0.09 ± 0.18 with one amplitude, mirroring the known shift between dust treatments in PR4 (Diego-Palazuelos et al. 2022).

5. IS THE CALIBRATION PATTERN SPATIALLY STATIONARY?

5.1. *Statistic*

We measure the 28 pair angles separately in R disjoint regions with covariances C_r (block-diagonal; map-level simulations give a mean cross-region correlation of $+0.002$, maximum 0.078, against a noise floor of 0.16). With $F_r = A^\top C_r^{-1} A$ ($A = d_0$ without the gauge column) the global-fit residual splits exactly,

$$\chi^2_{\rm total} = \sum_r \chi^2_{{\rm closure},r} + \sum_r (\hat\alpha_r - \hat\alpha)^\top F_r (\hat\alpha_r - \hat\alpha), \quad (2)$$

with $21R$ and $7(R-1)$ degrees of freedom. The second term, $\chi^2_{\rm spatial}$, tests whether the per-map pattern differs between regions. A sky-dependent *common* rotation, such as anisotropic birefringence, is gauge and invisible. The four same-frequency A/B edges compare detectors in the same band, so foregrounds cancel; variation on them is instrumental.

5.2. *Blinding and validation*

Validation used each region's real spectra with the antisymmetric part removed after derotating by the published global angles, so no regional pair angle was formed

before the analysis was frozen. Code hashes, statistic definitions and the threshold (PTE< 0.01; twelve HEALPix base pixels primary, eight latitude bands as a robustness partition) were committed before a single unblinding run. On 500 null skies the statistics match their degrees of freedom (mean $\chi^2_{\rm spatial} = 77.3$ for 77 dof; closure 252.5 for 252). The twelve-region test detects a 0.2° dust-like drift 92% and a 0.1° instrument-like drift 90% of the time.

5.3. *Result*

For twelve regions $\chi^2_{\rm spatial} = 70.9$ for 77 dof (PTE 0.67), closure 245.9 for 252 (PTE 0.60), A/B edges 45.8 for 44 (PTE 0.397) and no dependence on a 353 GHz dust tracer ($\Delta\chi^2 = 6.3$ for 7, PTE 0.50; Fig. 2). The calibration pattern is spatially stationary at this sensitivity. In the latitude partition $\chi^2_{\rm spatial} = 67.9$ for 49 (PTE 0.04) and the A/B edges give 49.0 for 28 (PTE 0.008), driven by the 143 GHz pair and the northern 20°–40° band in a pattern consistent with a shift of one map; over the eight statistics reported the Bonferroni-corrected value is 0.07. We report it as a target for split-based follow-up rather than a detection.

6. WHAT THE HFI CHANNELS SAY ABOUT β

6.1. *Simulation-based inference*

The MK estimate of β moves with the dust-EB model (Sect. 4). To quantify this without committing to one dust model, we train a neural posterior estimator. A bandpower simulator draws exact Wishart realizations of the 136 field spectra per bin from: the ΛCDM spectra (QuickPol windows, mask transfer) rotated by $\alpha_i + \beta$; a modified-blackbody dust component scaled from the measured 353 GHz spectra, whose polarization in band i is intrinsically rotated by a misalignment $\psi_i = \psi_0(\ell/200)^n + \psi_1 \ln(\nu_i/353\,\mathrm{GHz})$, so that the (E_i, B_j) dust block is $R(\psi_i)\,\mathrm{diag}(C^{EE}_{\rm dust}, C^{BB}_{\rm dust})\,R(\psi_j)^{\rm T}$ (for $\psi_i = \psi_j$ this is the familiar $C^{EB}_{\rm dust} = \frac{1}{2}\sin 4\psi\,(C^{EE}_{\rm dust} - C^{BB}_{\rm dust})$, and for $\psi_i \neq \psi_j$ it mimics a calibration difference); and the measured noise. This family contains no dust EB, a constant misalignment, scale dependence and frequency dependence. The angles α_i carry a broad common offset, $\mathcal{U}(-3°, 3°)$; a narrow one would let the prior, rather than the data, fix β. Summaries are the 14 scores at a fiducial point, and an ensemble of five conditional masked autoregressive flows is trained on 4×10^5 simulations. The network's log-densities are recomputed in float64 in the verification sandbox.

6.2. *Validation*

On 2,000 held-out simulations per dust model, the 68% and 95% intervals for β cover the truth 0.62 and 0.91 of the time for the full model (Fig. 3). The posterior is wide, $\sigma_\beta \simeq 0.63°$, and a Fisher analysis at the fiducial point shows why: $\sigma_\beta = 0.149°$ with no dust EB, but $\sigma_\beta = \infty$ once a constant misalignment is free. Dust appears rotated by $\alpha+\psi$ and the CMB by $\alpha+\beta$, so a common shift of α, β and ψ leaves the model unchanged. For a scale-independent misalignment ($n = 0$, the fiducial point) this is exact: the Fisher matrix has a single null direction, $\delta\alpha_i = -\epsilon$ for all i and $\delta\beta = \delta\psi_0 = \epsilon$ (relative eigenvalue $\sim 10^{-15}$). For $n \neq 0$ the same shift changes ψ by $\epsilon[(\ell/200)^n - 1]$, so only the scale dependence of the misalignment can break the degeneracy, and in the full model family the validated posterior for β is still prior-limited. The data constrain $\beta - \psi_0$ and $\alpha+\beta$, and β alone is set by the priors. Its marginal posterior is then a flat box bounded by prior edges, which the flows represent imperfectly; this is the origin of the mild under-coverage of β alone. The identifiable combination $\beta - \psi_0$ is recovered with $\sigma \simeq 0.34°$ and coverage 0.72 (68%) and 0.97 (95%). For β under an external prior on ψ_0, obtained by importance re-weighting, we test calibration on simulations drawn from that prior: for $\psi_0 \sim \mathcal{N}(0, 0.5°)$ the intervals cover 0.62 (68%) and 0.93 (95%), and for $\mathcal{N}(0, 1°)$ 0.61 and 0.93. Before unblinding we therefore fixed an empirical calibration: each reported interval is widened about its median by the factor that restores nominal coverage on these simulations (1.06 at 68% for $\mathcal{N}(0, 0.5°)$), never narrowed. For comparison, an MK fit without a dust-EB term covers the truth only 0.04 of the time when a constant misalignment is present, and even without dust EB its β error grows with the absolute common miscalibration (slope +0.18 per degree), an attenuation from noise in the observed C^{EE} that acts as its regressor.

6.3. *Result*

Applied once to the real spectra after a committed freeze, the posterior gives $\beta - \psi_0 = -0.09^{+0.29}_{-0.28}$ deg (68%, calibrated), consistent with our MK fit without dust EB, which measures the same combination. With no assumption on ψ_0, the β posterior returns the prior ($-0.01^{+0.83}_{-0.83}$ deg). Re-weighting to a prior $\psi_0 \sim \mathcal{N}(0, 0.5°)$ gives $\beta = -0.07^{+0.55}_{-0.52}$ deg. A precise β from these channels therefore requires an assumption about the mean dust misalignment, and the published shift between dust treatments (Diego-Palazuelos et al. 2022) is the expression of this degeneracy. Our $\beta - \psi_0$ lies 0.4° below the published no-dust-EB PR4 value of 0.30° (Diego-Palazuelos et al. 2022); we attribute this to pipeline differences (masks, pseudo-C_ℓ) that we cannot resolve without the original spectra.

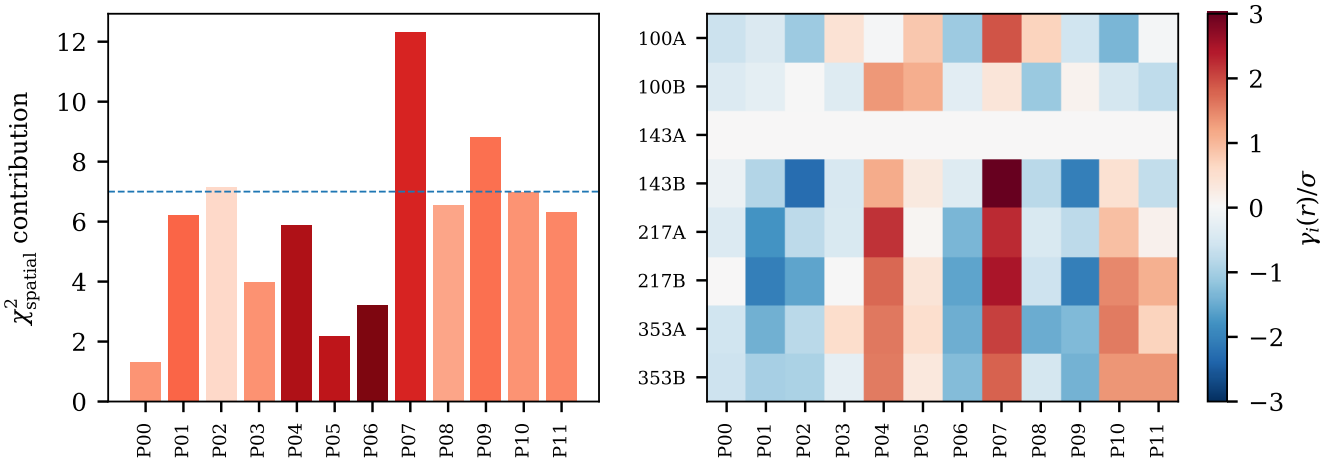


Figure 2. Left: per-region contributions to $\chi^2_{\rm spatial}$ (expected 7 each; darker bars carry more 353 GHz dust). Right: regional drift $\gamma_i(r)$ of each map in units of its error (gauge 143A).

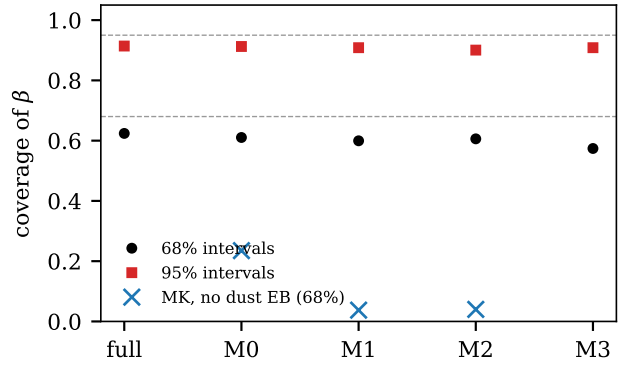


Figure 3. Coverage of β by the neural posterior for each dust-EB model, and by an MK fit without a dust-EB term (68%).

7. A TOPOLOGICAL DIAGNOSTIC OF THE TEMPERATURE MAP

Sublevel persistent homology (Cole & Shiu 2018) of three 20° patches of the 143 GHz $(A+B)/2$ temperature map (10′ smoothing, $\ell \leq 1000$) at the Galactic caps and the Cold Spot was compared with three null ensembles of 1,000 realizations each: isotropic Gaussian skies from the measured masked-sky spectrum; Gaussian CMB plus anisotropic white noise with Planck's per-pixel variance; and phase-randomized surrogates that keep the harmonic amplitudes of the masked map exactly. Monte Carlo p-values carry the +1 correction and are Holm-adjusted over all comparisons. The number of H_0 features is high against every null in all three patches: Gaussian $z = +2.90, +2.19, +2.62$ (NGP, SGP, Cold Spot); with anisotropic noise $+2.82, +2.08, +2.61$; and against the phase surrogates $+3.68, +3.00, +3.20$. The smallest Holm-adjusted p is 0.07, the floor set by the number of realizations. Building the surrogates from the unmasked map instead appears to remove the excess, but only because excluded Galactic-plane power then enters them; masking the real patches instead cuts compact-source holes into them that act as deep minima. None of these nulls reproduces the excess, though with Holm $p \geq 0.07$ none is excluded either. The leading candidate is the compact sources (and clusters) that the mask flags inside the patches and that no null contains; a source-masked or source-injected null is the next test. We report it as an unexplained diagnostic and claim no cosmological interpretation.

8. LIMITATIONS

Masks are rebuilt rather than identical to those of Lonappan et al. (2026); pseudo-C_ℓ replaces NaMaster deconvolution; Fisher scoring replaces MCMC; covariances are Gaussian. The spatial test is blind to anisotropic birefringence by construction, and regions are coarse (seven parameters per region). The A/B edge-constancy statistic treats the four same-frequency edges as independent within a region. A post-hoc check that keeps their full within-region covariance finds correlations $|\rho| \leq 0.024$ and leaves the statistics essentially unchanged (latitude bands $\chi^2 = 49.3$ for 28 dof instead of 49.0), but correlations between regions and uncertainty in the covariance itself are not modelled. None of the analyses here uses end-to-end NPIPE simulations.

9. CONCLUSIONS

The differential calibration network of Lonappan et al. (2026) reproduces on the public PR4 maps with an independent pipeline. Its cochain structure makes the gauge, the absence of harmonic modes and the equivalence of closure and fit χ^2 explicit, and it extends naturally to

a region-resolved test, under which the Planck calibration pattern is spatially stationary within the sensitivity stated above. A coverage-validated neural posterior shows that, with the dust misalignment free, β is degenerate with it, exactly so when the misalignment is scale-independent: these channels measure $\beta - \psi_0$, and any value of β quoted from them carries an assumption about ψ_0. Separating the two needs information that does not rotate with the dust, such as an absolute calibrator or a measured dust misalignment.

DATA AND CODE AVAILABILITY

Code, results, verification receipts and an interactive summary are public at https://github.com/ZuluYokohama/cmb-cst-lab. The MK likelihood is a port of the MIT-licensed pipeline of Lonappan et al. (2026). Planck PR4 data are available from NERSC and the Planck Legacy Archive.

Based on observations obtained with Planck, an ESA science mission. Analysis code was developed with the assistance of an AI coding assistant (Claude, Anthropic); the author is responsible for the analysis and its conclusions.

APPENDIX

A. VERIFICATION

Numerical kernels (network algebra, pair-angle estimator, MK likelihood, region statistics, bootstrap kernels and persistence) are numpy-only and executed in a network-less container with hash-chained receipts; GPU and NPU outputs are recomputed there in float64. Receipt identifiers are listed in the repository README.

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